

# Carl Bryant Carlos

BS Computer Science (AI), Mapua University

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Residency: National Capital Region, Philippines



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## CERTIFICATIONS AND AREAS OF INTEREST

- ❖ AWS Certified Solutions Architect – Associate
- ❖ AWS Certified AI Practitioner
- ❖ AWS Certified Cloud Practitioner
- ❖ TOEIC Score: 945 (CEFR C1)

Main interests: Networks, Cloud Computing, IT, DevOps, Artificial Intelligence, Infrastructure

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## TECHNICAL SKILLS

### Programming & Development

- Python, Java, C/C++, SQL, Bash, HTML/CSS
- **Machine Learning & AI:** PyTorch, Scikit-learn, LangChain, FAISS, NumPy, Pandas, deep learning, neural networks, CNN, few-shot learning, NLP, LLMs, RAG, model evaluation

### Cloud & DevOps

- **AWS:** VPC, subnets, EC2, S3, IAM, security groups, route tables, Lambda, RDS – architecting secure, scalable, and fault-tolerant distributed systems (SAA certified)
- **CI/CD:** Docker, GitHub Actions, Node-RED, n8n
- **Virtualization & Servers:** Nginx, Apache, VirtualBox, WSL, Proxmox
- **IoT & Edge Computing:** Raspberry Pi 5 (ARM64) and Arduino (Embedded Systems)

### Networking & Security

- **Protocols & Concepts:** OSI and TCP/IP, CGNAT, NAT, DNS, DHCP, MQTT, SNMP, LoRaWAN, HTTP/HTTPS, firewall, subnetting, tunneling, proxies
- **Tools:** Wireshark, Nmap/Zenmap, Netmonster, Cisco Packet Tracer, PRTG
- **VPN & Remote Access:** Wireguard, OpenVPN, MobaXterm, PuTTY, Termux, RealVNC

### Software Tools & Platforms

- **Development:** Visual Studio Code, Git/GitHub, Postman
  - **Productivity:** MS Office, Canva, Figma, AnyDesk, TeamViewer, Parsec
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## WORK EXPERIENCE

### IT Intern (R&D)

(Nov 2025 – Jan 2026 @ Geoplan Philippines, Inc.)

- Evaluated and configured IoT devices to assess product viability for PoC proposals and client deployments.
  - Assisted in on-site client PoC, integrated IoT devices into their PRTG monitoring platform via Node-RED workflows, and led troubleshooting of network connectivity issues.
  - Deployed a containerized IoT dashboard using Docker, integrating real-time device data ingestion via webhooks from LoRaWAN sensors.
  - Diagnosed a UWB Real-Time Location System's inconsistent connectivity and narrowed the root cause to protocol and cloud latency, prompting a server switch to on-premises testing.
  - Created technical documentation and tutorials in Google Slides to support internal knowledge base and client-facing presentations.
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## RESEARCH PAPERS AND PROJECTS HANDLED

### Binary Occupancy Detection: Utilizing Wi-BFI and Few-Shot Learning via Consumer Devices

*(Thesis, 2026 @ Mapúa University)*

- Detects whether a room is occupied or empty using only Wi-Fi; no cameras or sensors.
- Performed ETL on raw signal data from 3 consumer devices: cleaned, normalized, and structured into 5,729 labeled training samples.
- Trained a supervised CNN with Focal Loss and WeightedRandomSampler (94.9% PR-AUC).
- Trained a ProtoNet few-shot learner (98.0% PR-AUC at K = 10, 95% less labeling cost).
- Statistically validated as significant via one-sided Wilcoxon test with Bonferroni correction.

### TDA Classification using SVM for the Philippine Stock Exchange Index

*(Unpublished Research Activity for AI-2 Course, 2024 @ Mapúa University)*

- Classifies the market trend of the PSEI using topological data analysis and machine learning.
- Engineered features from raw PSEI time series data via Taken's embedding and Vietoris-Rips persistence homology (giotto-tda), then classified with an SVM (RBF kernel).
- The idea is "Point clouds tend to form clusters during market crashes."

### TReE: Triple Renewable Energy (Solar, Water, and Wind) as a Power Source of Weather Monitoring System

*(Research Project, 2022 @ Pasig City Science Highschool)*

- Handled the Arduino programming for the DHT22 sensor's readings display.
- Combines the three renewable energies into one generator.

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## TRAININGS

Full list (30+) at LinkedIn: <https://www.linkedin.com/in/carlcarlos>

|                                                              |         |
|--------------------------------------------------------------|---------|
| CCNA: Enterprise Networking, Security, and Automation, Cisco | – 08/25 |
| Foundations of Cybersecurity, Google                         | – 03/25 |
| Introduction to Cloud Computing, IBM                         | – 02/25 |
| Introduction to DevOps, IBM                                  | – 02/25 |
| BI Foundations with SQL, ETL and Data Warehousing, IBM       | – 05/24 |

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## EDUCATION

### Mapua University

Bachelor of Science in Computer Science with Specialization in Artificial Intelligence  
(2022 – 2026)  
GWA: 1.7820

### Pasig City Science Highschool

Science, Technology, Engineering, and Mathematics (2020 – 2021)

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## PERSONAL DATA

- Credly: <https://www.credly.com/users/carl-bryant-carlos>
- Full name: Carl Bryant Astacaan Carlos
- Nationality: Filipino